

Lecture (7): Blood supply of the brain

The brain is supplied by **two systems**:

Internal carotid system: formed of right & left internal carotid arteries.

Vertebrobasilar system: formed of right & left **vertebral** arteries and single **basilar** artery.

▪ Internal carotid artery:

➤ **Begin:** It originates as one of the two terminal branches of the common carotid artery at the upper border of thyroid cartilage opposite C4 vertebra.

➤ **Course:** it is divided into 4 parts:

1) **Cervical part:** inside the carotid sheath in the neck.

2) **Petrous part:** inside carotid canal in petrous temporal bone.

3) **Cavernous part:** inside cavernous sinus in cranial cavity. Gives superior and inferior hypophyseal arteries supply pituitary gland.

4) **Cerebral part:** also called supra-cavernous part.

➤ **Termination:** it ends lateral to optic chiasma, just below the anterior perforated substance of the brain by dividing into two terminal branches; anterior and middle cerebral arteries.

➤ **Branches of the Cerebral part:**

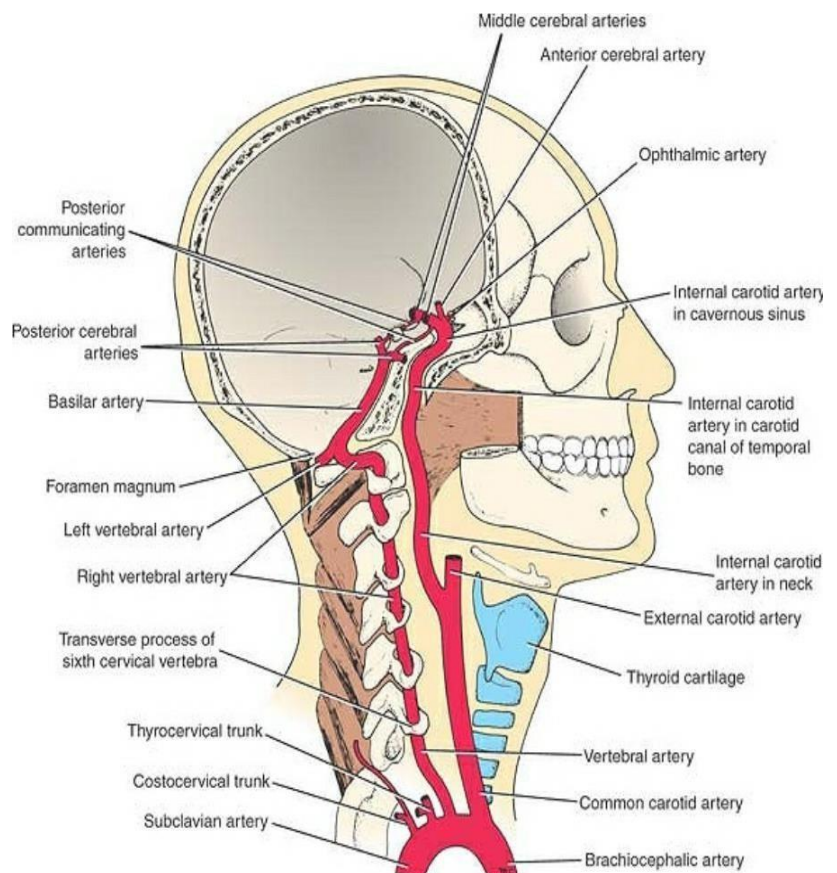
1. **Ophthalmic artery** which enters the orbit through optic canal.

2. **The posterior communicating artery.**

3. The **anterior choroidal artery** supplies the posterior limb of internal capsule, choroid plexuses of the 3rd & lateral ventricle.

4. **Anterior cerebral artery.**

5. **Middle cerebral artery.**



▪ Anterior cerebral artery:

➤ **Origin:** it is the smaller of the two terminal branches of internal carotid artery.

➤ **Course:**

- It passes forwards and medially towards the longitudinal fissure between the two cerebral hemispheres.
- It joins with its fellow by anterior communicating artery.
- It reaches the genu of corpus callosum, then runs backwards in callosal sulcus on the upper surface of corpus callosum.

➤ **Termination:** it ends at the splenium of corpus callosum by terminal branches.

➤ **Branches:**

1- **Cortical branches:**

- Medial half of orbital surface of cerebral hemisphere.
- Upper one inch of the superolateral surface of cerebral hemisphere till parieto-occipital sulcus.

2- **Callosal branches:** to supply all parts of corpus callosum except splenium.

3- **Central branches:** penetrate anterior perforated substance to supply:

- Anterior part of corpus striatum.
- Anterior part of anterior limb of internal capsule.
- Anterior parts of hypothalamus and septum pellucidum.

➤ **Clinical importance:**

- It supplies **motor and sensory areas** of the lower limb in the **paracentral lobule**. Occlusion of anterior cerebral artery causes **contralateral somatosensory loss** and **contralateral paralysis** of the leg.
- Affection of corpus callosum causes **apraxia** which is inability to perform purposeful movements.

▪ Middle cerebral artery:

➤ **Origin:** It is the larger of the two terminal branches of internal carotid artery.

➤ **Course:** It passes laterally in the stem of lateral sulcus, then it passes backwards in the posterior ramus of lateral sulcus over the insula.

➤ **Termination:** It ends over the **insula** by dividing into many terminal branches which appear on the lateral surface of cerebral hemisphere.

➤ **Branches:**

1. **Cortical branches:**

- The lateral half of the orbital surface of the cerebral hemisphere. ○ Insula and temporal pole.
- The superolateral surface of cerebral hemisphere except upper one inch, lower one

inch and occipital lobe.

2. Central branches (lateral striate arteries) supply:

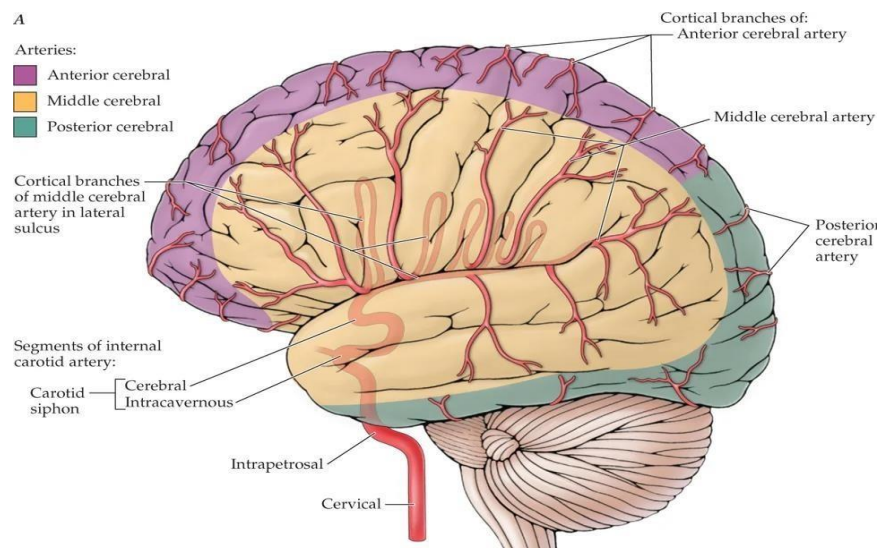
- Posterior part of corpus striatum.
- Anterior limb, genu and posterior limb of internal capsule.
- One of these lateral striate arteries is large and liable to rupture called **artery of cerebral hemorrhage**.

➤ Clinical importance:

- It supplies the motor and sensory areas of the whole body except leg area (lower limb).
- It supplies auditory areas in the temporal lobe.
- It supplies motor speech area (Broca's area) in frontal lobe and sensory speech area in parietal lobe.

➤ Middle cerebral artery syndrome (occlusion of MCA):

- Contralateral sensory loss of the arm and leg.
- Contralateral paralysis of the arm and leg.
- Lesion to **area 8** leads to Ipsilateral eye deviation.
- Broca (motor) **aphasia** and Wernicke's (receptive) aphasia.
- **Partial deafness** is due to lesions to auditory cortex.

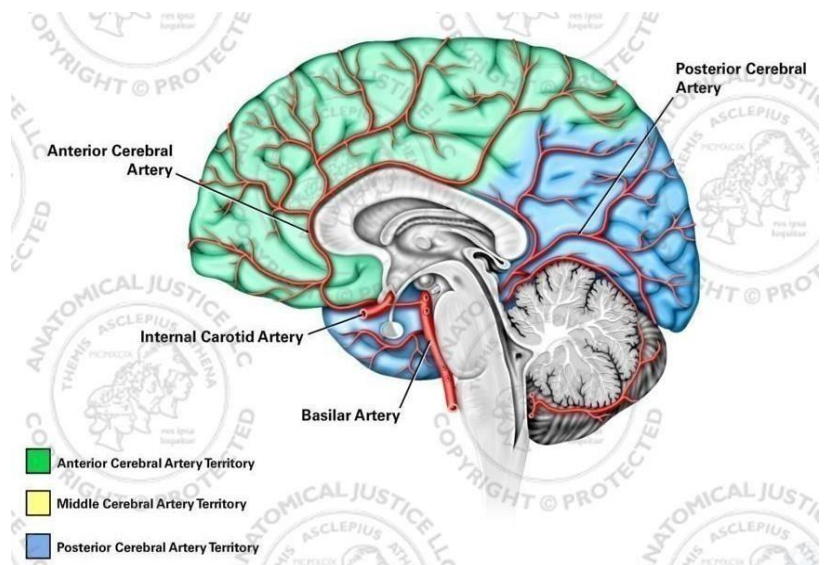


■ Posterior cerebral artery:

- **Origin:** Right and left posterior cerebral arteries are the two terminal branches of the basilar artery.

➤ Course:

- It curves laterally and backwards around the cerebral peduncles of midbrain.
- It reaches below the splenium where it divides into many



temporal branches.

- It passes in calcarine sulcus.

➤ **Termination:** it ends by dividing into parietooccipital and calcarine branches.

➤ **Branches:**

1- **Cortical branches:** It supplies

- The **inferior (tentorial) surface** of cerebral hemisphere except **temporal pole**.

- The **lower one inch** of the superolateral surface of the cerebral hemisphere.

- **All aspects of the occipital lobe.**

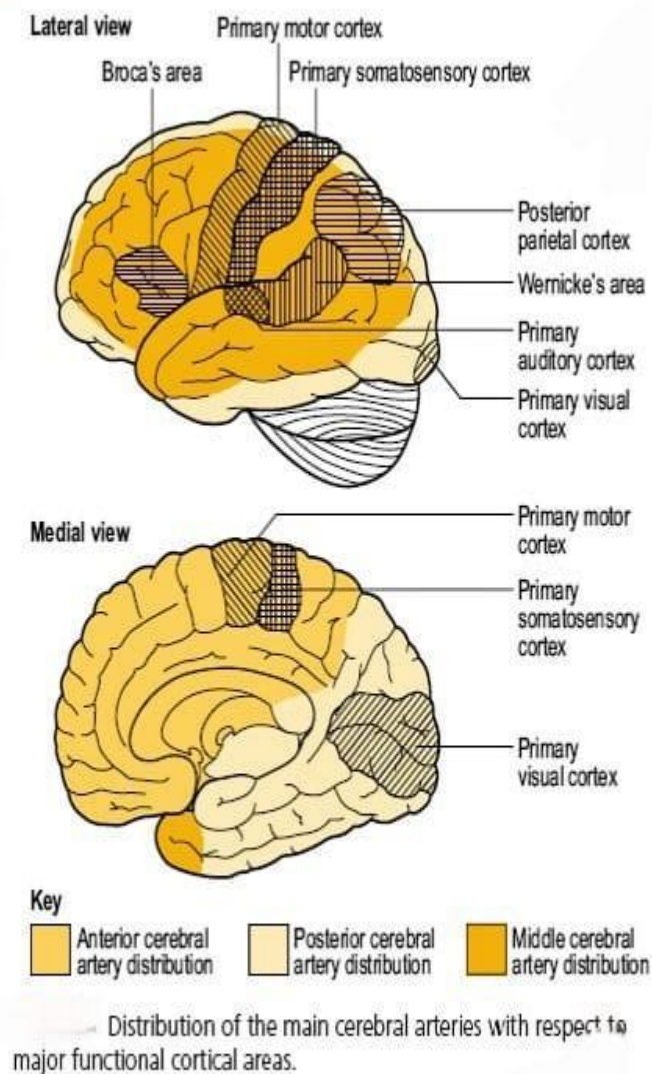
- The **occipital visual cortex** and **incus (Centre of smell)**.

2- **Callosal branches** supply the splenium of corpus callosum.

3- **Central branches** are divided into short medial group supplies the cerebral peduncles of midbrain & the mammillary bodies; long lateral group supplies the geniculate bodies and thalamus.

4- **Posterior choroidal artery** supplies the thalamus and the choroidal plexus of the third and lateral ventricles.

➤ **Lesion to this artery** causes contralateral homonymous hemianopia with macular sparing and visual agnosia. Midbrain lesion leads to contralateral hemiplegia with lateral



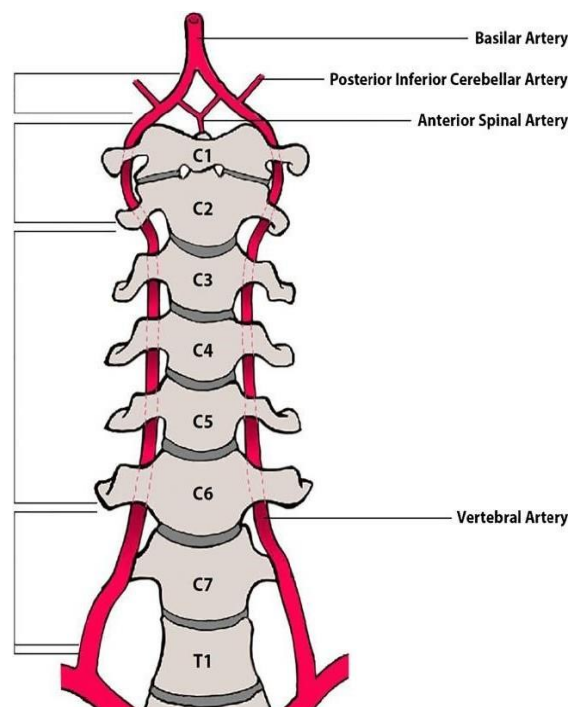
Functional areas associated with the main cerebral arteries

Cerebral artery	Cortical area
Anterior cerebral	Paracentral lobule: primary motor cortex and primary somatosensory cortex regions (hip to feet) Supplementary motor area (movement) Prefrontal and orbitofrontal cortex (cognition and emotion) Corpus callosum Septal nucleus (pleasure)
Middle cerebral	Frontal lobe: primary motor cortex (hip to head), premotor area, frontal eye field Parietal cortex: primary somatosensory cortex (hip to head), primary taste cortex Temporal lobe: primary auditory cortex (hearing), primary olfactory cortex (smell) Basal ganglia (movement initiation) Optic radiation (vision) Uncus (emotion) Dominant hemisphere: language centres—Wernicke's area (receptive speech) and Broca's area (expressive speech) Non-dominant hemisphere: contralateral awareness of self and surroundings
Posterior cerebral	Visual cortex (primary and association) Hippocampus (long-term memory) Thalamus Hypothalamus (autonomic function)

squint, ataxia and tremors.

■ Vertebral artery:

- **Origin:** it arises from the first part of subclavian artery.
- **Course:**
 - 1st part lies medial to scalenus anterior muscle in neck.
 - 2nd part ascends through the foramina transversaria of the upper 6 cervical vertebrae.
 - 3rd part in suboccipital triangle.
 - 4th part enters the cranial cavity through the



foramen magnum in front of medulla oblongata.

- **Termination:** It ends by uniting with its fellow to form the basilar artery at the lower border of pons.
- **Branches: (4th part)**
 - 1- **Anterior spinal artery supplies medulla & spinal cord.**
 - 2- **Rt & Lt posterior spinal arteries.**
 - 3- **Medullary branches.**
 - 4- **Posterior inferior cerebellar artery (PICA)** which supplies the inferior cerebellar peduncle, lateral part of medulla and posterior part of inferior surface of cerebellum.
- **Occlusion of anterior spinal artery** causes **medial medullary syndrome** which includes lesion of pyramidal tract & hypoglossal nucleus results in contralateral hemiplegia & ipsilateral paralysis of tongue (hypoglossal alternating hemiplegia).
- **Occlusion of posterior inferior cerebellar artery causes lateral medullary syndrome (Wallenburg's syndrome) resulting in:**
 - Cerebellar ataxia due to lesion of inferior cerebellar peduncle.
 - Contralateral loss of pain sensation from the body due to lesion of spinal lemniscus.
 - Ipsilateral loss of pain sensation from the face due to lesion of spinal nucleus of trigeminal.
 - Ipsilateral paralysis of muscles of palate, pharynx and larynx.

Basilar artery:

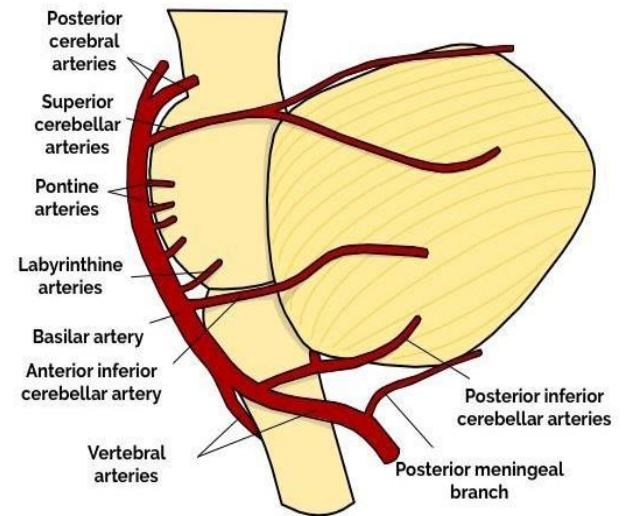
- **Origin:** It is formed by the union of **two vertebral arteries** at the lower border of pons.
- **Course:** it ascends in basilar groove on the ventral surface of the pons.
- **Termination:** It ends by dividing into two terminal branches (Rt & Lt posterior cerebral arteries). at the upper border of the pons.
- **Branches:**
 - **Medial and Lateral pontine Branches** supply the pons.
 - **Anterior inferior cerebellar artery (AICA)** to the anterior part of the inferior surface of the cerebellum.
 - **Labyrinthine artery** to the inner ear.
 - **Superior cerebellar artery** supplies superior surface of the cerebellum.
 - **Posterior cerebral artery** (two terminal branches).

- **Occlusion of Medial pontine branches**
cause **medial pontine syndrome**

(**Millard-Gubler syndrome**) which means lesion of pyramidal tract & facial nucleus, contralateral hemiplegia & ipsilateral facial lesion (facial alternating hemiplegia).

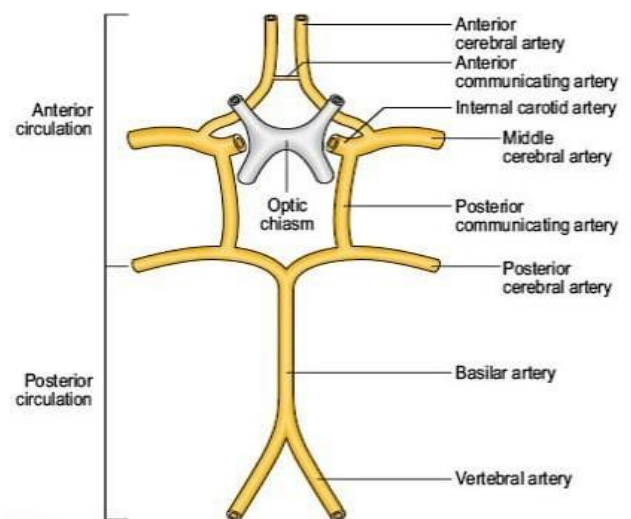
- **Occlusion of lateral pontine branches cause Lateral pontine syndrome** lesion

of middle cerebellar peduncle & cerebellar ataxia.



Circle of Willis (Circulus Arteriosus):

- The internal carotid and vertebrobasilar systems anastomose with each other around the **optic chiasm** and **infundibulum** of the pituitary stalk in the base of the forebrain.
- Communicating vessels allow equalization of blood flow between the two sides of the brain and can allow collateral circulation if parts are occluded.
- The circle is formed of **nine branches** as follows:



Schematic view of the circle of Willis.

- **Rt & Lt posterior cerebral arteries (PCA).**
- **Rt & Lt posterior communicating arteries between ICA & PCA.**
- **Rt & Lt internal carotid arteries (ICA).**
- **Rt & Lt anterior cerebral arteries (ACA).**